**MetaSUB: Metadata protocol (surface sampling)**

*In this guide, you will find an explanation of the of the online data collection tool KoBoToobox (https://kf.kobotoolbox.org) used for data collection by the MetaSUB Consortium.*


**I. METADATA……2**

**II. KOBOTOOLBOX……2**

i. Collection Survey
ii. Registration & Logging in
iii. Accessing the project
iv. Sharing the project
v. Submitting entries
    i. Offline entries/Draft
vi. Following data collection

**III. DATA COLLECTION (BY HAND/EXCEL) ……9**

**IV. COLLECTION SURVEY OUTLINE……..…10**

**V. CONTACT……..……..18**

## METADATA

Due to the large scale of most MetaSUB projects, a key goal of the MetaSUB Consortium is a requirement for detailed practices and protocols. This includes sampling methods, standardization of protocols for data collection and processing, and validation and comparability of metadata. Standardization of metadata for all MetaSUB cities is pivotal for efficient characterization and analysis of samples as a whole. It is also vital for proper and complete metadata is attached to each sample taken.

Dr. Christopher Mason once said, "*A sample is only as good as it's metadata*", which is a principle the MetaSUB Consortium stands by. Any sample without metadata cannot be processed.

The minimum metadata needed for each sample is as follows: barcode, sample type (control vs. experiment), location, geolocation (latitude, longitude), timepoint, surface material, object sampled (i.e. ticket machine, handrail).

## KOBOTOOLBOX

KoBoToolbox is a free open-source survey software for mobile data collection, compatible with most modern devices browsers. The MetaSUB Consortium has used KoBoToolbox for gCSD (Global City Sampling Day) since 2017.



KoBoToolbox

https://kf.kobotoolbox.org

Cities participating in gCSD are encouraged to use KoBoToolbox. All metadata points collected on KoBoToolbox are stored on an online server where it can be secure and easily accessed. Submitted entries are stored in your browser cache, and when you turn on the internet, they will be sent to the KoBoToolbox server.

Samples collected can be seen on the MetaSUB website map for each sampling year.

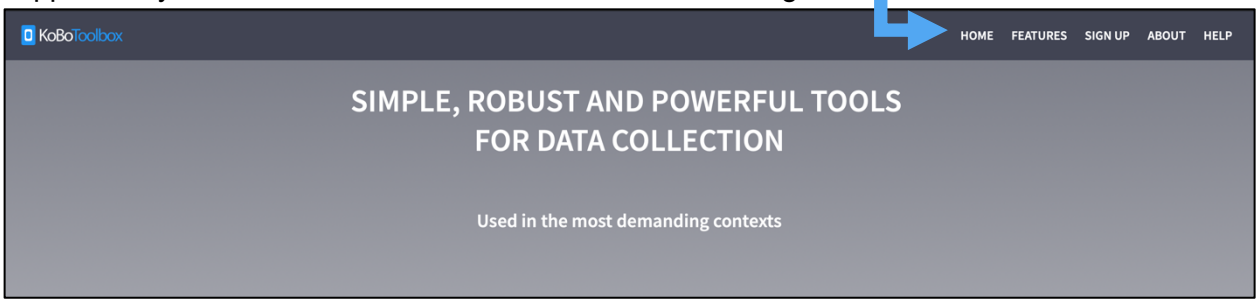
# REGISTRATION & LOGGING IN

- **Registration form: https://kf.kobotoolbox.org/accounts/register/**

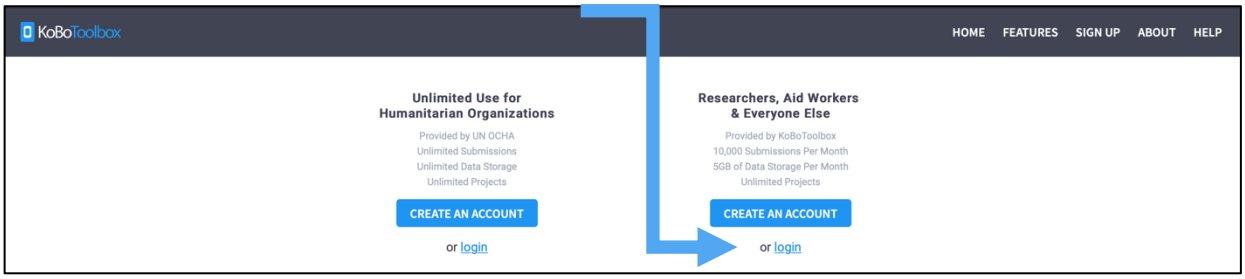
**Follow the directions to register. Use small case for login username only.**
KoBoToolbox will then send you a **confirmation link** through your email account, which you **must follow** in order to complete the registration.

- **Logging in**

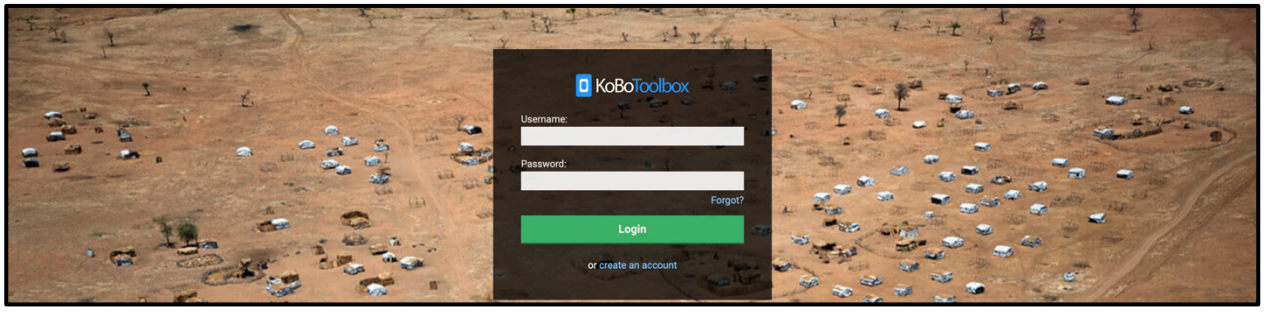
Use the link (https://www.kobotoolbox.org) to access KoBoToolbox on a computer. It is supported by most browsers. Click SIGN UP in order to find login.




Once you proceed. Click **login** under **Researchers, Aid Workers & Everyone Else** to find the login page.




Once at the following page you can login to KoBoToolbox and access the collection form.

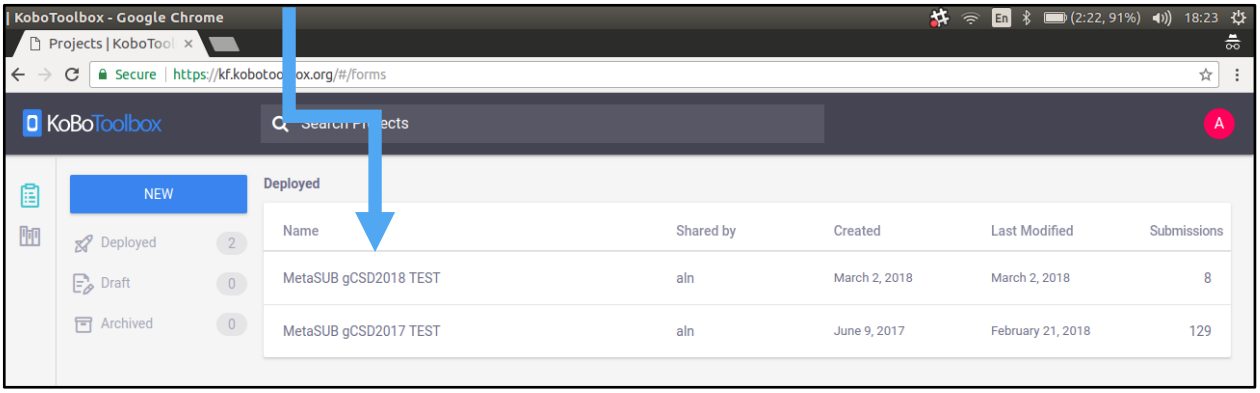


# ACCESSING THE PROJECT

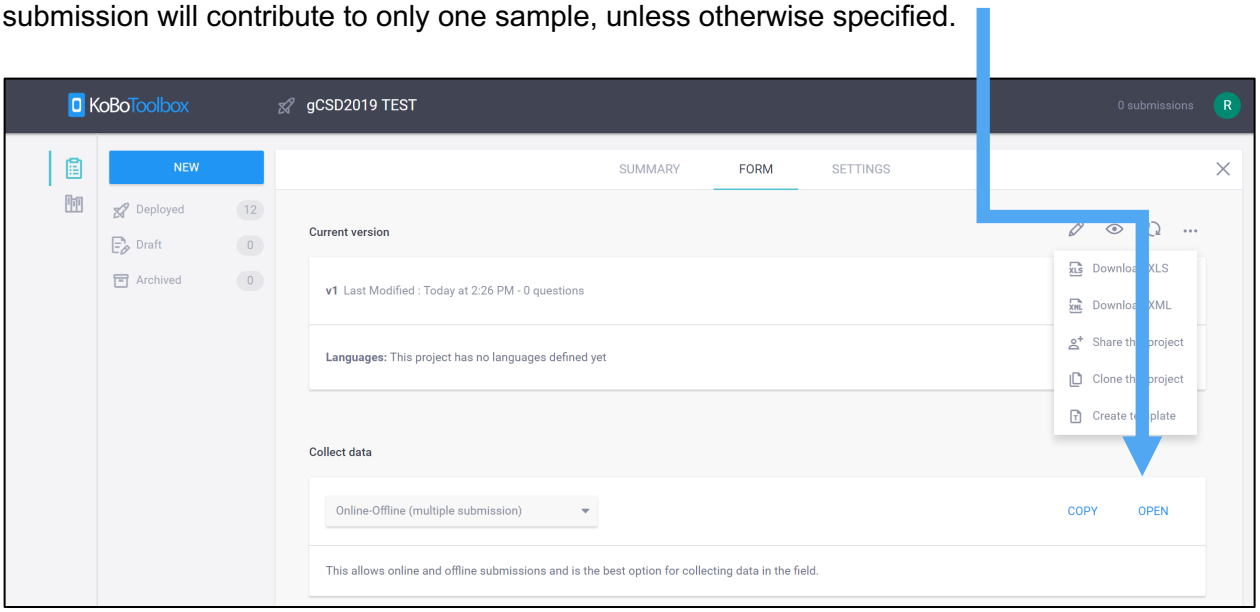
Log into your KoBoToolbox Account.

Register if you have not done so, following the instructions above. Please **send your username** to Krista Ryon (krr4004@med.cornell.edu) once registered.

Use https://kf.kobotoolbox.org/accounts/login/ to login and see the project shared with your account, under 'Deployed'.



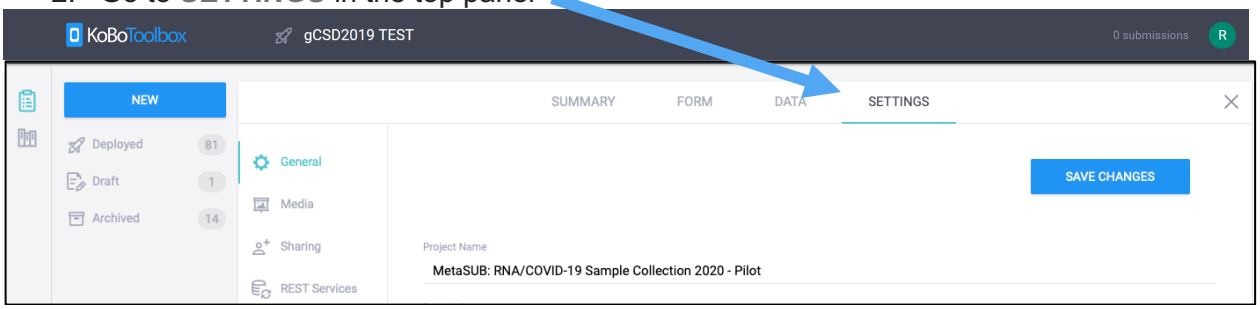

Click the form and you will see the screen below. In order to start a 'submission' you need to click **OPEN**. You will then be able to start entering data for one microbe. Please note, each submission will contribute to only one sample, unless otherwise specified.



# SHARING THE PROJECT

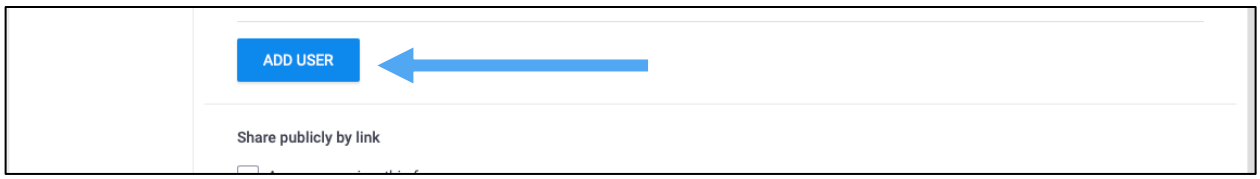
KoBoToolbox projects can be shared to multiple users. Specific collaborators, team leaders, will be able to add more users to their projects.

1. Access the project
2. Go to **SETTINGS** in the top panel



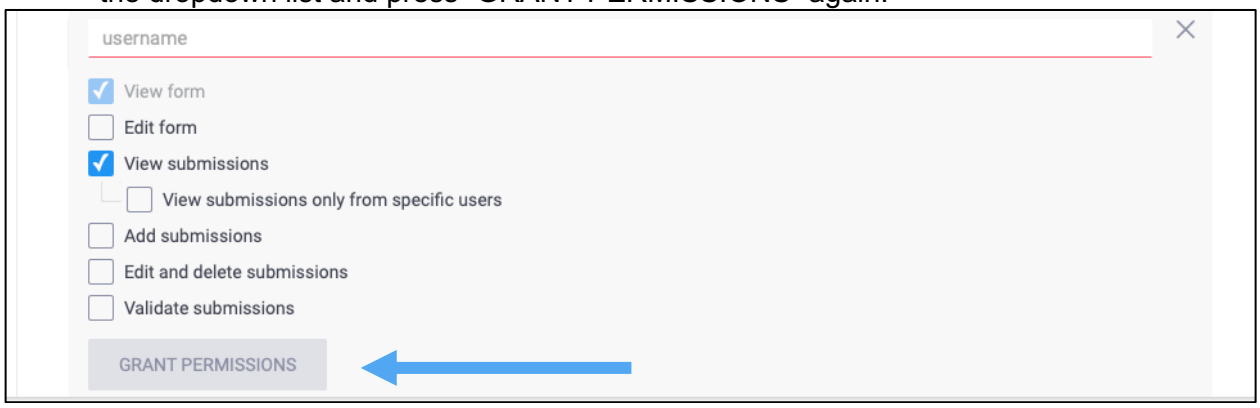

3. On the left panel, click Sharing 

and then 'ADD USER'



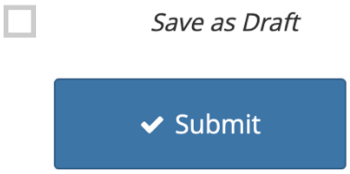

4. Input the username of the person you wish to share the form with and click 'Add submissions'. KoBoToolbox now has the option to allow specific users to view submissions.

   Add Submissions" permission is enough to be able to add new entries into the collection form. If you want to add a user with full permissions, you need to assign multiple permissions to the same user. Just input username again, choose next permission from the dropdown list and press "GRANT PERMISSIONS" again.



# SUBMITTING ENTRIES

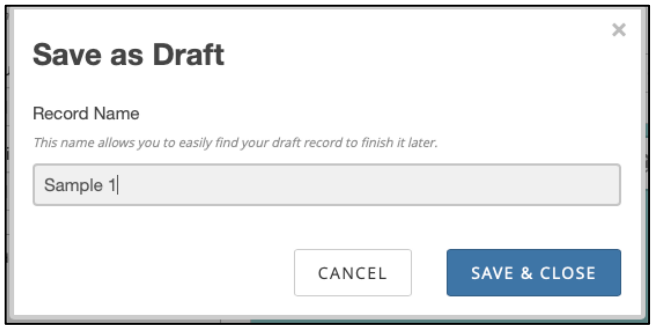
After data collection, you can either submit data or save as a draft.




**Submit:** this will save your collected data immediately; this requires access to the internet.

**Save as Draft:** this will store the collected data in your device (mobile phone etc.) until you have access to reliable cellular or internet services.

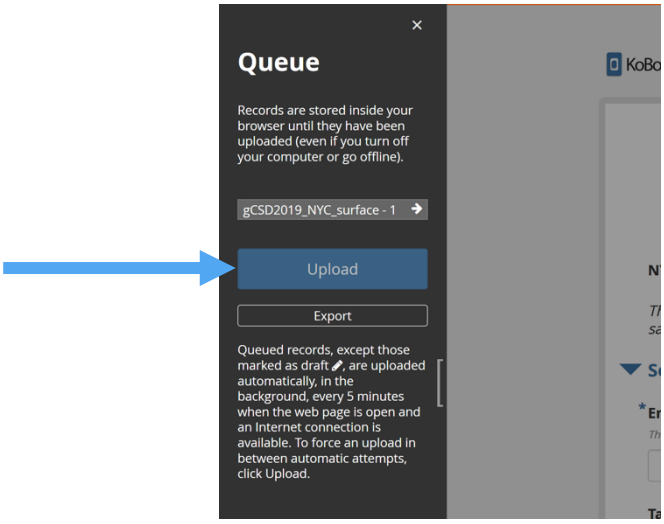
After you **Save as Draft** you will be prompted to name the record and save




Once you save, on the top of the form there will be an orange graph icon. The number beside it represents the number of saved drafts.




Once you are ready to submit your drafts, click on the orange icon. A sidebar will pop-up and show your queued drafts.



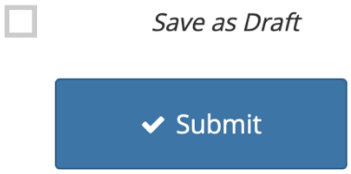

You can queue more than one draft while sampling if needed.
*We recommend submitting the forms as soon as you have access to the internet.*

**There are three options in Queue:**

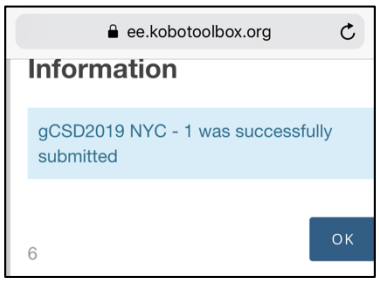
1. Click on the arrow to pull your draft back into the form. This can be useful for incomplete forms.

2. Click **'Upload'** to submit your form and remove it from drafts. Note that the form will upload complete drafts automatically every 5 minutes when connected to the internet and your browser is open.

3. Click **'Export'** to download your submission directly.




**Submit:** this will save your collected data immediately; this requires access to the internet.

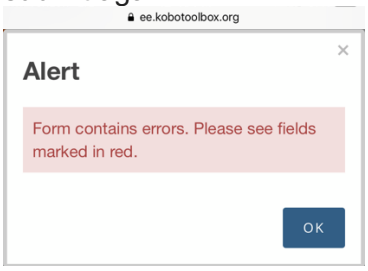



Once submitted, you record will be queued until you see a prompt indicating your form was successfully submitted.

This is similar to the draft, but there will be no option to save your submission

**Error**

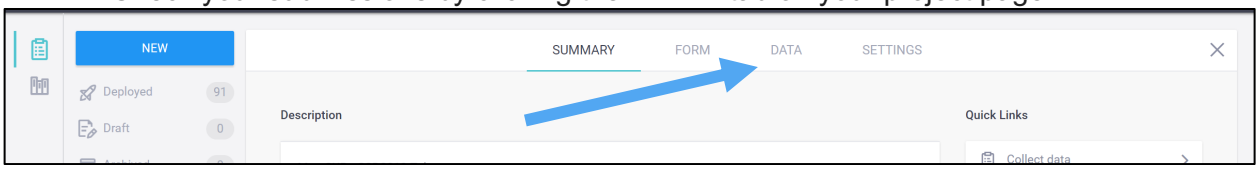
If this appears than your form contains errors. The fields/questions that contain the errors will be marked in red. Return to the form and correct the error before trying to submit again.



# FOLLOWING DATA COLLECTION

Once sampling is complete you will be able to view all your cities submission and data. This permission is given to the project leads in each city.

Check your submissions by clicking the ‘DATA’ tab on your project page.




Click on ‘Table’ to view each submission individually. Please verify that all your team members have submitted their forms (if saved on as a draft) on their device (phone, iPad etc.) and that the number of records here matches the sample and test samples collected.

The project’s data will be downloaded following your project by the **metasub_admin**. Please verify that all the collected metadata matches your records following sampling day.

# DATA COLLECTION BY HAND

We recommend taking notes on each of your samples if there are any errors. This is a great source for data cross-checking if we run into any problems.

A paper form can be designed if requested for your city.

If your city is planning on entering data solely by hand and sending the metadata after in an online sheet, please email Benjamin Young (bgy2102@columbia.edu) with your data collection plan.

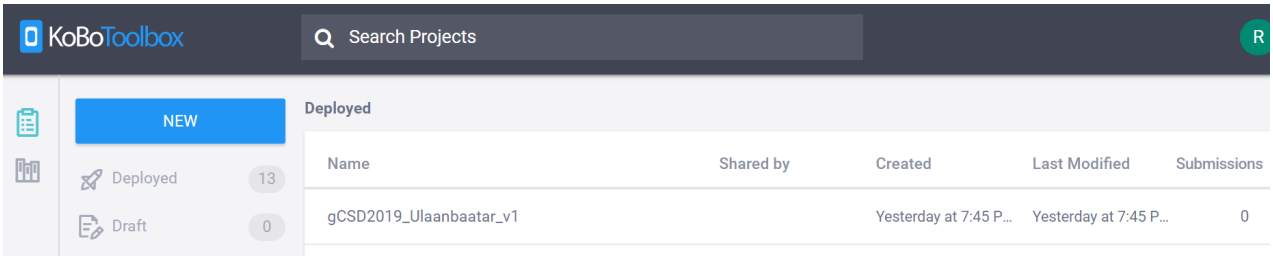
If you choose not to use KoBoToolbox, please send your metadata to Benjamin Young and Krista Ryon in an excel spreadsheet.

Metadata Template.xls

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Barcode** | **Station** | **Sampling place** | **Surface Material** | **Comment** |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |
| 11 | | | | | |
| 12 | | | | | |
| 13 | | | | | |

# COLLECTION SURVEY OUTLINE

Access your city-specific survey through your KoBoToolbox homepage. If you do not see your survey, make sure you have shared your KoBoToolbox username with krr4004@med.cornell.edu. If you have used KoboToolbox previously for gCSD, we will email you to confirm your user.
*Forms will be shared 1 week prior to gCSD

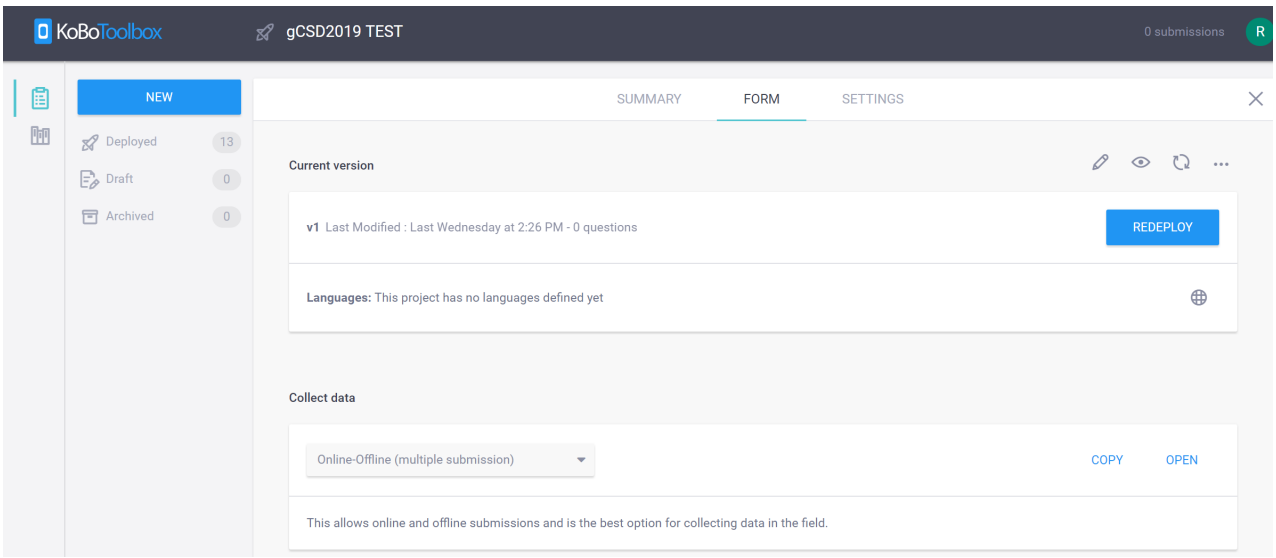



Click on the name of the survey to access:

**The name should be called gCSD2021 'Your City Name'**

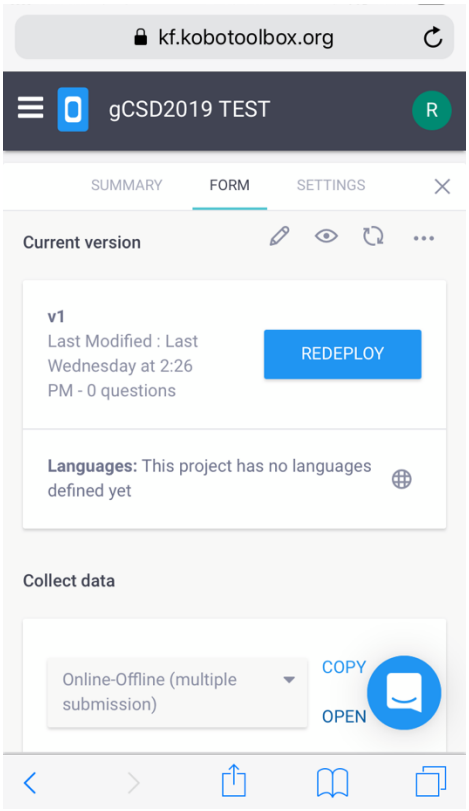
*(ex. gCSD2021 Paris)*

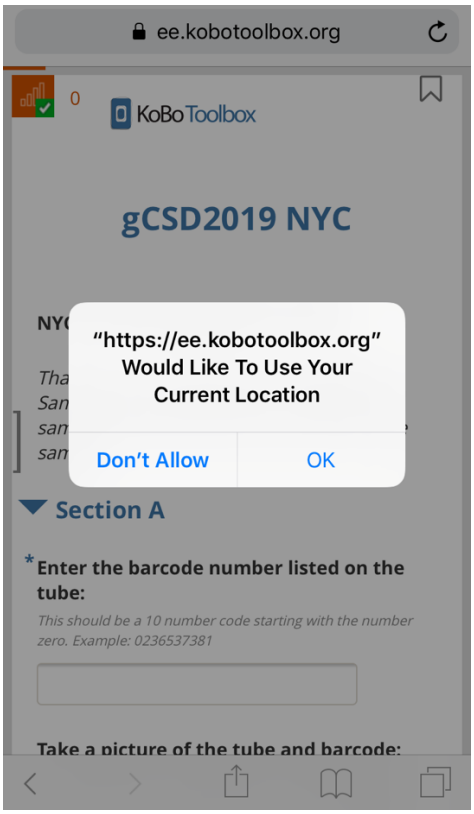
To start collection, press 'OPEN' under the tab 'FORM' and section 'Collect data'



**Note: We recommend that you collect data in KoBoToolbox using a mobile smartphone browser (i.e. Safari, Google Chrome, Firefox)**

When prompted, click 'OK' in order for the browser to use your location and capture your GPS data.




## QUESTIONS

| | |
|---|---|
| **CITY & COUNTRY**<br>The name of your city and country will be prefilled unless you are sampling multiple cities. | City<br>Ulaanbaatar<br>Country<br>Mongolia |
| **SAMPLING TYPE (SURFACE)**<br>**Note that *unless otherwise specified*, you will be collecting a 'surface' sample (as opposed to air, water, soil etc.)** | |

## BARCODE NUMBER

Enter your barcode information located on the side of the sample tube under the barcode. **This will be a 10 digit code.**

**If a code is entered that is more or less than 10 digits you will be unable to submit the survey,**



* **Enter the barcode number listed on the tube:**

*This should be a 10 number code starting with the number zero. Example: 0236537381*

* **Enter the barcode number listed on the tube:**

*This should be a 10 number code starting with the number zero. Example: 0236537381*

038383

**Value not allowed**

## PICTURE/PHOTO OF BARCODE

**Please take a photo of the tube and barcode prior to sampling. Make sure the barcode is visible in the photo with the digits.**

Press the text box and you will be prompted to take a photo or browse your device.

ee.kobotoolbox.org

**Take a picture of the tube and barcode:**

Click here to upload file. (< 10MB)

## SAMPLE TYPE

Select either **Experiment**, **Negative Control** or **Positive Control.**

If 'Negative Control' is selected, questions regarding sampling place or surface material will not appear.

* **Select sample type**

*Experiment, or Positive Control, or Negative Control*

- Experiment
- Negative Control
- Positive Control

## LOCATION TYPE

**Location type will be specific to each cities sampling plan.**

A general format will be provided without a sampling plan.

** Following location type you will be prompted to answer:

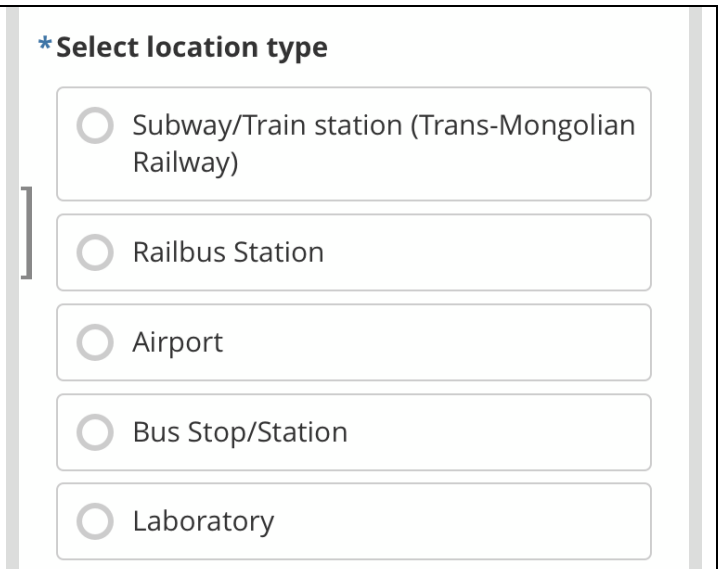
**Line** (line or route the location is on)

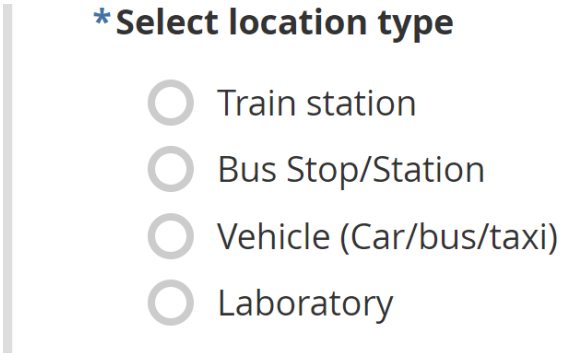
**Location** ( name of the location )

**Sublocation** (additional location information)

Again, depending on your cities unique sampling plan these options may change.

**If you chose 'Laboratory' as your location type, these options will not appear**




- **LOCATION NAME**

The name of your location, with specific details that can pinpoint the location if there is no geolocation attached to the sample.

- **LINE**

If you are sampling at a station, please specify the 'line' or 'route' the transportation runs.

- **SUBLOCATION**

Additional information about the location.

## SETTING

**Urban:** relating to a city that is generally very developed with a density of human structures

**Suburban:** an area that is mixed-used or a residential area that has a semi-dense population.

**Rural:** relating to an area that may have open land and a lower population density when compared to an urban environment. There may be more land between structures and settlements.

***Setting**

- ○ Urban
- ○ Suburban
- ○ Rural

## GROUNDLEVEL

**Underground (or underwater):** an area underground such as an underground metro/subway or an area built below ground level

**Aboveground (open area):** an area aboveground that is exposed to the outdoor environment

**Aboveground (closed area):** an area aboveground that is closed from the outdoor environment (indoors), such as inside a building or station.

**Aboveground (mixed area):** an area that is aboveground that is both open and closed. There is some exposure to the outdoor environment.

***Select ground level**

- ○ Underground (or underwater)
- ○ Aboveground (open area)
- ○ Aboveground (closed area)
- ○ Aboveground (mixed area)

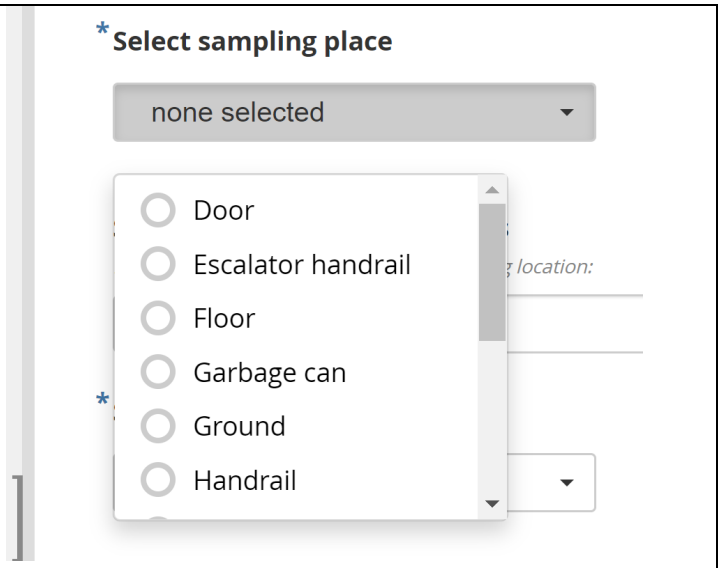
## OBJECT TYPE / SAMPLING PLACE

Options will be determined by existing metadata and information from the cities sampling plan.

Please include any specific information on the location.

Some options may include but are not limited to:

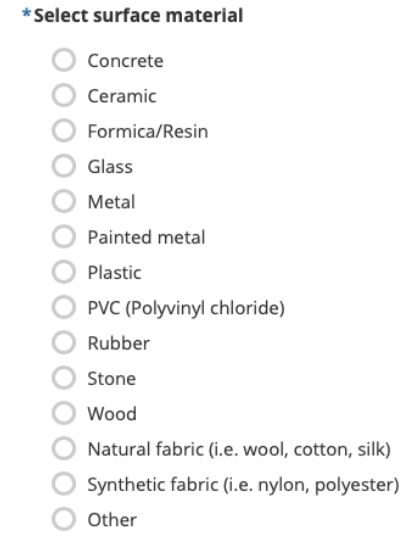
- Door
- Escalator handrail
- Floor
- Garbage can
- Ground
- Handrail
- Turnstile
- Ticket machine
- Seat
- Bench




## SURFACE MATERIAL

Select the material that best describes the item swabbed.

When sampling, swab within one type of surface material. Example, a ticket machine may be comprised of plastic and metal. Stay within one surface type.



## TRAFFIC

Select how many people are currently at your location while collecting your sample.

This is a general approximation.

***Traffic at the sampling location:**

*Approximately how many people are currently at the location*

- ○ 0
- ○ 1 - 10
- ○ 11 - 20
- ○ 21 - 30
- ○ 31 - 40
- ○ 41 - 50
- ○ 50+

## TEMPERATURE & HUMIDITY

If you are not collecting temperature or humidity you can press 'No' and continue the form. Air temperature and relative humidity can be measured unless otherwise specified (if you have the capability to measure surface temperature, dew point etc.)

***Was temperature measured at this sampling location?**

*Yes or No*

- ◉ Yes
- ○ No

**Input average air temperature:**

*°Celsius*

***Was humidity measured at this sampling location?**

*Yes or No*

- ◉ Yes
- ○ No

**Input Relative humidity**

*%*

## GEOLOCATION

**Geolocation can be entered after collection if locations and geolocations are provided to the MetaSUB planning team.**

**Geolocation for a mobile device:**

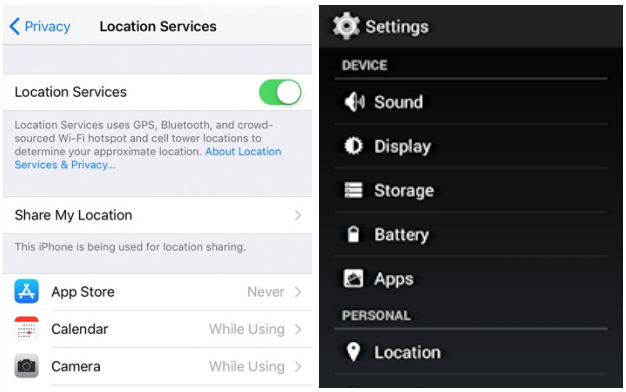
- Press the button in order to capture your geolocation
- If you know your geolocation you can enter the latitude (x, y°), longitude (x, y°), and altitude (m) manually.

If you are still experiencing issues still, enter a 0.1 into each geolocation field and in 'additional comments' please input a GPS error.




Note, if you are unable to collect your GPS data check if your browser's location settings are turned on in your device setting.

This will be different depending on the mobile device you are using and its OS.



### TIMEPOINT & DATE

If you are submitting the sample information on another day or at another time, please enter the date and time the sample was taken.

KoBoToolbox will automatically collect date and time if you are collecting samples and submitting metadata simultaneously.

### COMMENTS

If you have any additional information regarding the sample, please enter it into the comments box.

***Examples***

GPS Error

Test sample

Sources of error

**Enter any additional comments if necessary:**

*For example, possible sources of error, weather condition etc.*

**Note:** Please submit all metadata for gCSD by September 1st unless you have notified the MetaSUB leadership of any additional sampling.

## CONTACT INFO

Collaborators, please email the MetaSUB leadership with **any issues during your sample collection** and remember to **exercise recommended safety precautions at all times**.

**Krista Ryon, Executive Director MetaSUB**
krr4004@med.cornell.edu